KERALA DIPLOMA CURRICULUM REVISION 2026

Course 4121 | 4 Credits: 4 | Program Core | Source-grounded

Wood Products I

 To familiarize students with veneer manufacturing process in plywood industry,

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4121 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4121.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	4121
Course title	Wood Products I
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

14 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To familiarize students with veneer manufacturing process in plywood industry,
- possibility of solid and laminated wood,and wood bending processes

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding production Summaries veneer processing and manufacture	See official syllabus
CO2	14 Understanding of plywood Interpret grading of plywood , testing and	See official syllabus
CO3	15 Applying properties Compare solid wood bending and laminated wood	See official syllabus
CO4	14 Applying bending process	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the raw materials used for plywood production	4	As specified
Module 2	Summarise veneer processing and manufacture of plywood	4	As specified
Module 3	Interpret grading of plywood, testing and properties	3	As specified
Module 4	Compare solid wood bending and laminated wood bending process	3	As specified

MODULE 1

Outline the raw materials used for plywood production

This module is presented from the official syllabus scope for Wood Products I. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the raw materials used for plywood production
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 1 Remembering manufacturing Explain Processing of Logs Prior to

1 Remembering manufacturing Explain Processing of Logs Prior to is an official topic in Module 1 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Remembering manufacturing Explain Processing of Logs Prior to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 remembering manufacturing explain processing of logs prior to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 1 Remembering manufacturing Explain Processing of Logs Prior to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 1 Remembering manufacturing Explain Processing of Logs Prior to □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 5 Understanding Veneering

5 Understanding Veneering is an official topic in Module 1 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Veneering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding veneering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Veneering means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 5 Understanding Veneering □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Illustrate rotary cutting and Splicing of veneer

Illustrate rotary cutting and Splicing of veneer is an official topic in Module 1 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate rotary cutting and Splicing of veneer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate rotary cutting and splicing of veneer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate rotary cutting and Splicing of veneer means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Illustrate rotary cutting and Splicing of veneer □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Outline Drying and Splicing of Veneer 5 Applying Plywood, veneer, flitches, peeling, slicing and drying - advantage of plywood over solid wood - history of plywood manufacture - factors considering in the selection of logs and its storage and transport - end coating and end protection devices - softening treatment of logs prior to boiling and slicing - flitches and their preparation - tangential and radial cutting - veneer slicing and methods - vertical slicing machine - rotary peeling - nose bar, knife arrangement on peeling lathe - Illustrate schematic diagram of log with respect to knife and nose bar - handling of veneers and clipping - advantage of veneer drying - Jet dryer, roller dryer, band dryer, breather and Kiln dryer - processing of veneers - storage and handling of veneers from peeling - joining and splicing machine- cross feed splicer

Outline Drying and Splicing of Veneer 5 Applying Plywood, veneer, flitches, peeling, slicing and drying - advantage of plywood over solid wood - history of plywood manufacture - factors considering in the selection of logs and its storage and transport - end coating and end protection devices - softening treatment of logs prior to boiling and slicing - flitches and their preparation - tangential and radial cutting - veneer slicing and methods - vertical slicing machine - rotary peeling - nose bar, knife arrangement on peeling lathe - Illustrate schematic diagram of log with respect to knife and nose bar - handling of veneers and clipping - advantage of veneer drying - Jet dryer, roller dryer, band dryer, breather and Kiln dryer - processing of veneers - storage and handling of veneers from peeling - joining and splicing machine- cross feed splicer is an official topic in Module 1 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Drying and Splicing of Veneer 5 Applying Plywood, veneer, flitches, peeling, slicing and drying - advantage of plywood over solid wood - history of plywood manufacture - factors considering in the selection of logs and its storage and transport - end coating and end protection devices - softening treatment of logs prior to boiling and slicing - flitches and their preparation - tangential and radial cutting - veneer slicing and methods - vertical slicing machine - rotary peeling - nose bar, knife arrangement on peeling lathe - Illustrate schematic diagram of log with respect to knife and nose bar - handling of veneers and clipping - advantage of veneer drying - Jet dryer, roller dryer, band dryer, breather and Kiln dryer - processing of veneers - storage and handling of veneers from peeling - joining and splicing machine- cross feed splicer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline drying and splicing of veneer 5 applying plywood, veneer, flitches, peeling, slicing and drying - advantage of plywood over solid wood - history of plywood manufacture - factors considering in the selection of logs and its storage and transport - end coating and end protection devices - softening treatment of logs prior to boiling and slicing - flitches and their preparation - tangential and radial cutting - veneer slicing and methods - vertical slicing machine - rotary peeling - nose bar, knife arrangement on peeling lathe - illustrate schematic diagram of log with respect to knife and nose bar - handling of veneers and clipping - advantage of veneer drying - jet dryer, roller dryer, band dryer, breather and kiln dryer - processing of veneers - storage and handling of veneers from peeling - joining and splicing machine- cross feed splicer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Drying and Splicing of Veneer 5 Applying Plywood, veneer, flitches, peeling, slicing and drying - advantage of plywood over solid wood - history of plywood manufacture - factors considering in the selection of logs and its storage and transport - end coating and end protection devices - softening treatment of logs prior to boiling and slicing - flitches and their preparation - tangential and radial cutting - veneer slicing and methods - vertical slicing machine - rotary peeling - nose bar, knife arrangement on peeling lathe - Illustrate schematic diagram of log with respect to knife and nose bar - handling of veneers and clipping - advantage of veneer drying - Jet dryer, roller dryer, band dryer, breather and Kiln dryer - processing of veneers - storage and handling of veneers from peeling - joining and splicing machine- cross feed splicer means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Outline Drying and Splicing of Veneer 5 Applying Plywood, veneer, flitches, peeling, slicing and drying - advantage of plywood over solid wood - history of plywood manufacture - factors considering in the selection of logs and its storage and transport - end coating and end protection devices - softening treatment of logs prior to boiling and slicing - flitches and their preparation - tangential and radial cutting - veneer slicing and methods - vertical slicing machine - rotary peeling - nose bar, knife arrangement on peeling lathe - Illustrate schematic diagram of log with respect to knife and nose bar - handling of veneers and clipping - advantage of veneer drying - Jet dryer, roller dryer, band dryer, breather and Kiln dryer - processing of veneers - storage and handling of veneers from peeling - joining and splicing machine- cross feed splicer □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Summarise veneer processing and manufacture of plywood

This module is presented from the official syllabus scope for Wood Products I. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise veneer processing and manufacture of plywood
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▾ Illustrate repairing of Veneer

Illustrate repairing of Veneer is an official topic in Module 2 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate repairing of Veneer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate repairing of veneer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate repairing of Veneer means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate repairing of Veneer □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ Illustrate Veneer matching

Illustrate Veneer matching is an official topic in Module 2 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Veneer matching using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate veneer matching should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate Veneer matching means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate Veneer matching □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Explain gluing of veneer

Explain gluing of veneer is an official topic in Module 2 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain gluing of veneer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain gluing of veneer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain gluing of veneer means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Explain gluing of veneer □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Outline assembling of veneer 4 Understanding Repair of veneers - picus joint, scarf joint, Ibus joint - veneer matching - four piece matching, book matching side to side matching, end to end matching cathedral matching - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - pre pressing and its advantages - cold press - hot press - gauge pressure - conditioning - hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps - dimensioning and sanding of plywood - trimming - standard plywood specifications - drum sander, belt sander, wide belt sander - abrasive materials

Outline assembling of veneer 4 Understanding Repair of veneers - picus joint, scarf joint, Ibus joint - veneer matching - four piece matching, book matching side to side matching, end to end matching cathedral matching - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - pre pressing and its advantages - cold press - hot press - gauge pressure - conditioning - hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps - dimensioning and sanding of plywood - trimming - standard plywood specifications - drum sander, belt sander, wide belt sander - abrasive materials is an official topic in Module 2 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline assembling of veneer 4 Understanding Repair of veneers - picus joint, scarf joint, Ibus joint - veneer matching - four piece matching, book matching side to side matching, end to end matching cathedral matching - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - pre pressing and its advantages - cold press - hot press - gauge pressure - conditioning - hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps - dimensioning and sanding of plywood - trimming - standard plywood specifications - drum sander, belt sander, wide belt sander - abrasive materials using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline assembling of veneer 4 understanding repair of veneers - picus joint, scarf joint, ibus joint - veneer matching - four piece matching, book matching side to side matching, end to end matching cathedral matching - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - pre pressing and its advantages - cold press - hot press - gauge pressure - conditioning - hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps - dimensioning and sanding of plywood - trimming - standard plywood specifications - drum sander, belt sander, wide belt sander - abrasive materials should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline assembling of veneer 4 Understanding Repair of veneers - picus joint, scarf joint, Ibus joint - veneer matching - four piece matching, book matching side to side matching, end to end matching cathedral matching - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - pre pressing and its advantages - cold press - hot press - gauge pressure - conditioning - hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps - dimensioning and sanding of plywood - trimming - standard plywood specifications - drum sander, belt sander, wide belt sander - abrasive materials means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline assembling of veneer 4 Understanding Repair of veneers - picus joint, scarf joint, Ibus joint - veneer matching - four piece matching, book matching side to side matching, end to end matching cathedral matching - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - pre pressing and its advantages - cold press - hot press - gauge pressure - conditioning - hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps - dimensioning and sanding of plywood -

trimming - standard plywood specifications - drum sander, belt sander, wide belt sander - abrasive materials □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Interpret grading of plywood, testing and properties

This module is presented from the official syllabus scope for Wood Products I. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Interpret grading of plywood, testing and properties
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▾ Illustrate Pressing and Pre-Pressing

Illustrate Pressing and Pre-Pressing is an official topic in Module 3 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate Pressing and Pre-Pressing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate pressing and pre-pressing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate Pressing and Pre-Pressing means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Illustrate Pressing and Pre-Pressing □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▼ Explain Dimensioning of Plywood

Explain Dimensioning of Plywood is an official topic in Module 3 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Dimensioning of Plywood using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain dimensioning of plywood should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Dimensioning of Plywood means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain Dimensioning of Plywood □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Outline Grading of Plywood 5 Understanding Grading of Plywoods Physical properties - density, moisture - content, hygroscopic properties, absorption, swelling and shrinkage Mechanical properties - elasticity, rigidity tensile strength, bending Thermal conductivity and acoustical properties Testing of plywood as per BIS code - determination of density, moisture content - glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity Speciality plywoods, Tea chest plywood, marine plywood, shuttering plywood, fire proof plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded Plywood.

Outline Grading of Plywood 5 Understanding Grading of Plywoods Physical properties - density, moisture - content, hygroscopic properties, absorption, swelling and shrinkage Mechanical properties - elasticity, rigidity tensile strength, bending Thermal conductivity and acoustical properties Testing of plywood as per BIS code - determination of density, moisture content - glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity Speciality plywoods, Tea chest plywood, marine plywood, shuttering plywood, fire proof plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded Plywood. is an official topic in Module 3 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline Grading of Plywood 5 Understanding Grading of Plywoods Physical properties - density, moisture - content, hygroscopic properties, absorption, swelling and shrinkage Mechanical properties - elasticity, rigidity tensile strength, bending Thermal conductivity and acoustical properties Testing of plywood as per BIS code - determination of density, moisture content - glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity Speciality plywoods, Tea chest plywood, marine plywood, shuttering plywood, fire proof plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded Plywood. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline grading of plywood 5 understanding grading of plywoods physical properties - density, moisture - content, hygroscopic properties, absorption, swelling and shrinkage mechanical properties - elasticity, rigidity tensile strength, bending thermal conductivity and acoustical properties testing of plywood as per bis code - determination of density, moisture content - glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity speciality plywoods, tea chest plywood, marine plywood, shuttering plywood, fire proof plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded plywood. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline Grading of Plywood 5 Understanding Grading of Plywoods Physical properties - density, moisture - content, hygroscopic properties, absorption, swelling and shrinkage Mechanical properties - elasticity, rigidity tensile strength, bending Thermal conductivity and acoustical properties Testing of plywood as per BIS code - determination of density, moisture content - glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity Speciality plywoods, Tea chest plywood, marine plywood, shuttering plywood, fire proof plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded Plywood. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Outline Grading of Plywood 5 Understanding Grading of Plywoods Physical properties - density, moisture - content, hygroscopic properties, absorption, swelling and shrinkage Mechanical properties - elasticity, rigidity tensile strength, bending Thermal conductivity and acoustical properties Testing of plywood as per BIS code - determination of density, moisture content - glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity Speciality plywoods, Tea chest plywood, marine plywood, shuttering plywood, fire proof plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded Plywood.

□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Compare solid wood bending and laminated wood bending process

This module is presented from the official syllabus scope for Wood Products I. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Compare solid wood bending and laminated wood bending process
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Recall Wood Bending

Recall Wood Bending is an official topic in Module 4 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Recall Wood Bending using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, recall wood bending should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Recall Wood Bending means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Recall Wood Bending □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Explain Solid Wood Bending

Explain Solid Wood Bending is an official topic in Module 4 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Solid Wood Bending using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain solid wood bending should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Solid Wood Bending means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain Solid Wood Bending □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Outline laminated wood bending 5 Applying Bending - General Consideration in bending - steaming - cold bending and Hot Bending - simple ‘U’ bend with figure - Re entrant and ‘S’ type bend with figure - end stops - chair leg bending - Machine Bending - Machine bending - Rope and windless Machine with figure - Lever arm Machine with figure - Revolving Table Machine - setting technique - Movement of Bends - laminated bending - selection and Preparation of laminates - Pre bending - laminated wood bending by male & female forms - split female forms, by inflated flexible hose - two plane bends Curved laminated Members- Movement and distortion of laminated bends- Plywood bending Text / Reference: T/R Book Title/Author F.P. Kollmann, Principles of wood science and technology Vol.I, II - Springer T1 Verlag R1 W.C. Stevens ,Solid and laminated wood bending - Majesty’s stationary office R2 Lecturer note on plywood technology Vol.I, II - IPIRI, Bangalore. Online Resources: Sl.No Website Link 1 https://youtu.be/o7YnSV8P-IQ 2 https://www.youtube.com/watch?v=MzDaQ8z3jII 3 https://www.youtube.com/watch?v=cuMJ6c5r7UQ

Outline laminated wood bending 5 Applying Bending - General Consideration in bending - steaming - cold bending and Hot Bending - simple ‘U’ bend with figure - Re entrant and ‘S’ type bend with figure - end stops - chair leg bending - Machine Bending - Machine bending - Rope and windless Machine with figure - Lever arm Machine with figure - Revolving Table Machine - setting technique - Movement of Bends - laminated bending - selection and Preparation of laminates - Pre bending - laminated wood bending by male & female forms - split female forms, by inflated flexible hose - two plane bends Curved laminated Members- Movement and distortion of laminated bends- Plywood bending Text / Reference: T/R Book Title/Author F.P. Kollmann, Principles of wood science and technology Vol.I, II - Springer T1 Verlag R1 W.C. Stevens ,Solid and laminated wood bending - Majesty’s stationary office R2 Lecturer note on plywood technology Vol.I, II - IPIRI, Bangalore. Online Resources: Sl.No Website Link 1 https://youtu.be/o7YnSV8P-IQ 2 https://www.youtube.com/watch?v=MzDaQ8z3jII 3 https://www.youtube.com/watch?v=cuMJ6c5r7UQ is an official topic in Module 4 of Wood Products I. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline laminated wood bending 5 Applying Bending - General Consideration in bending - steaming - cold bending and Hot Bending - simple ‘U’ bend with figure - Re entrant and ‘S’ type bend with figure - end stops - chair leg bending - Machine Bending - Machine bending - Rope and windless Machine with figure - Lever arm Machine with figure - Revolving Table Machine - setting technique - Movement of Bends - laminated bending - selection and Preparation of laminates - Pre bending - laminated wood bending by male & female forms - split female forms, by inflated flexible hose - two plane bends Curved laminated Members- Movement and distortion of laminated bends- Plywood bending Text / Reference: T/R Book Title/Author F.P. Kollmann, Principles of wood science and technology Vol.I, II - Springer T1 Verlag R1 W.C. Stevens ,Solid and laminated wood bending - Majesty’s stationary office R2 Lecturer note on plywood technology Vol.I, II - IPIRI, Bangalore. Online Resources: Sl.No Website Link 1 https://youtu.be/o7YnSV8P-IQ 2 https://www.youtube.com/watch?v=MzDaQ8z3jII 3 https://www.youtube.com/watch?v=cuMJ6c5r7UQ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline laminated wood bending 5 applying bending - general consideration in bending - steaming - cold bending and hot bending - simple ‘u’ bend with figure - re entrant and ‘s’ type bend with figure - end stops - chair leg bending - machine bending - machine bending - rope and windless machine with figure - lever arm machine with figure - revolving table machine - setting technique - movement of bends - laminated bending - selection and preparation of laminates - pre bending - laminated wood bending by male & female forms - split female forms, by inflated flexible hose - two plane bends curved laminated members- movement and distortion of laminated bends- plywood bending text / reference: t/r book title/author f.p. kollmann, principles of wood science and technology vol.i, ii - springer t1 verlag r1 w.c. stevens ,solid and laminated wood bending - majesty’s stationary office r2 lecturer note on plywood technology vol.i, ii - ipiri, bangalore. online resources: sl.no website link 1 https://youtu.be/o7ynsv8p-iq 2 https://www.youtube.com/watch?v=mzdaq8z3jii 3 https://www.youtube.com/watch?v=cumj6c5r7uq should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline laminated wood bending 5 Applying Bending - General Consideration in bending - steaming - cold bending and Hot Bending - simple ‘U’ bend with figure - Re entrant and ‘S’ type bend with figure - end stops - chair leg bending - Machine Bending - Machine bending - Rope and windless Machine with figure - Lever arm Machine with figure - Revolving Table Machine - setting technique - Movement of Bends - laminated bending - selection and Preparation of laminates - Pre bending - laminated wood bending by male & female forms - split female forms, by inflated flexible hose - two plane bends Curved laminated Members- Movement and distortion of laminated bends- Plywood bending Text / Reference: T/R Book Title/Author F.P. Kollmann, Principles of wood science and technology Vol.I, II - Springer T1 Verlag R1 W.C. Stevens ,Solid and laminated wood bending - Majesty’s stationary office R2 Lecturer note on plywood technology Vol.I, II - IPIRI, Bangalore. Online Resources: Sl.No Website Link 1 https://youtu.be/o7YnSV8P-IQ 2 https://www.youtube.com/watch?

Aspect	What to prepare
	v=MzDaQ8z3jII 3 https://www.youtube.com/watch?v=cuMJ6c5r7UQ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline laminated wood bending 5 Applying Bending - General Consideration in bending - steaming - cold bending and Hot Bending - simple ‘U’ bend with figure - Re entrant and ‘S’ type bend with figure - end stops - chair leg bending - Machine Bending - Machine bending - Rope and windless Machine with figure - Lever arm Machine with figure - Revolving Table Machine - setting technique - Movement of Bends - laminated bending - selection and Preparation of laminates - Pre bending - laminated wood bending by male & female forms - split female forms, by inflated flexible hose - two plane bends Curved laminated Members- Movement and distortion of laminated bends- Plywood bending Text / Reference: T/R Book Title/Author F.P. Kollmann, Principles of wood science and technology Vol.I, II - Springer T1 Verlag R1 W.C. Stevens ,Solid and laminated wood bending - Majesty’s stationary office R2 Lecturer note on plywood technology Vol.I, II - IPIRI, Bangalore. Online Resources: Sl.No Website Link 1 https://youtu.be/o7YnSV8P-IQ 2 https://www.youtube.com/watch?v=MzDaQ8z3jII 3 https://www.youtube.com/watch?v=cuMJ6c5r7UQ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

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The official assessment structure is reproduced from the supplied PDF.
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> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 1 Remembering manufacturing Explain Processing of Logs Prior to. (3 marks)**

▶ **Define or explain 5 Understanding Veneering. (3 marks)**

▶ **Define or explain Illustrate rotary cutting and Splicing of veneer. (3 marks)**

▶ **Define or explain Outline Drying and Splicing of Veneer 5 Applying Plywood, veneer, flitches, peeling, slicing and drying - advantage of plywood over solid wood - history of plywood manufacture - factors considering in the selection of logs and its storage and transport - end coating and end protection devices - softening treatment of logs prior to boiling and slicing - flitches and their preparation - tangential and radial cutting - veneer slicing and methods - vertical slicing machine - rotary peeling - nose bar, knife arrangement on peeling lathe - Illustrate schematic diagram of log with respect to knife and nose bar - handling of veneers and clipping - advantage of veneer drying - Jet dryer, roller dryer, band dryer, breather and Kiln dryer - processing of veneers - storage and handling of veneers from peeling - joining and splicing machine- cross feed splicer. (3 marks)**

Module 2

▶ **Define or explain Illustrate repairing of Veneer. (3 marks)**

▶ **Define or explain Illustrate Veneer matching. (3 marks)**

▶ **Define or explain Explain gluing of veneer. (3 marks)**

▶ **Define or explain Outline assembling of veneer 4 Understanding Repair of veneers - picus joint, scarf joint, Ibus joint - veneer matching - four piece matching, book matching side to side matching, end to end matching cathedral matching - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - pre pressing and its advantages - cold press - hot press - gauge pressure - conditioning - hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps - dimensioning and sanding of plywood - trimming - standard plywood specifications - drum sander, belt sander, wide belt sander - abrasive materials. (3 marks)**

Module 3

▶ **Define or explain Illustrate Pressing and Pre-Pressing. (3 marks)**

▶ **Define or explain Explain Dimensioning of Plywood. (3 marks)**

▶ **Define or explain Outline Grading of Plywood 5 Understanding Grading of Plywoods Physical properties - density, moisture - content, hygroscopic properties, absorption, swelling and shrinkage Mechanical properties - elasticity, rigidity tensile strength, bending Thermal conductivity and acoustical properties Testing of plywood as per BIS code - determination of density, moisture content - glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity Speciality plywoods, Tea chest plywood, marine plywood, shuttering plywood, fire proof**

plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded Plywood.. (3 marks)

Module 4

▶ **Define or explain Recall Wood Bending. (3 marks)**

▶ **Define or explain Explain Solid Wood Bending. (3 marks)**

▶ **Define or explain Outline laminated wood bending 5 Applying Bending - General Consideration in bending - steaming - cold bending and Hot Bending - simple 'U' bend with figure - Re entrant and 'S' type bend with figure - end stops - chair leg bending - Machine Bending - Machine bending - Rope and windless Machine with figure - Lever arm Machine with figure - Revolving Table Machine - setting technique - Movement of Bends - laminated bending - selection and Preparation of laminates - Pre bending - laminated wood bending by male & female forms - split female forms, by inflated flexible hose - two plane bends Curved laminated Members- Movement and distortion of laminated bends- Plywood bending Text / Reference: T/R Book Title/Author F.P. Kollmann, Principles of wood science and technology Vol.I, II - Springer T1 Verlag R1 W.C. Stevens ,Solid and laminated wood bending - Majesty's stationary office R2 Lecturer note on plywood technology Vol.I, II - IPIRI, Bangalore. Online Resources: Sl.No Website Link 1 https://youtu.be/o7YnSV8P-IQ 2 https://www.youtube.com/watch?v=MzDaQ8z3jII 3 https://www.youtube.com/watch?v=cuMJ6c5r7UQ. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Remembering manufacturing Explain Processing of Logs Prior to**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Illustrate repairing of Veneer**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate Pressing and Pre-Pressing**.

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Recall Wood Bending**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://youtu.be/o7YnSV8P-IQ https://www.youtube.com/watch?v=MzDaQ8z3jlI https://www.youtube.com/watch?v=cuMJ6c5r7UQ

IN-PAGE SEARCH

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Prepared from the supplied official SITTTR syllabus PDF for Course 4121. Verify institutional updates against the current official curriculum.